

Experiment 7: Code Injection on Application Process using Ptrace (TryHackMe)

Aim

To demonstrate code injection on a running Linux application process using ptrace concepts in the TryHackMe platform.

Procedure

1. Login to TryHackMe and start a Linux-based room.
2. Launch AttackBox or connect via SSH.
3. Identify running processes.
4. Attach to process using ptrace-based tools.
5. Inspect registers and memory.
6. Modify execution flow conceptually.
7. Resume process execution.

Commands Used

```
ps aux  
strace -p <PID>  
gdb -p <PID>
```

Output

The target process was successfully traced and manipulated.

Result

Thus, code injection using ptrace concepts was demonstrated in TryHackMe.

Experiment 8: Privilege Escalation in TryHackMe Platform

Aim

To escalate privileges from a low-level user to root in a Linux system using TryHackMe.

Procedure

1. Start Linux PrivEsc room.
2. Gain low-privilege shell.
3. Enumerate system details.
4. Identify misconfigurations.
5. Exploit vulnerability.
6. Verify root access.

Commands Used

```
whoami  
id  
uname -a  
sudo -l  
find / -perm -4000 2>/dev/null
```

Output

Root access obtained successfully.

Result

Thus, privilege escalation was successfully performed in TryHackMe.

Experiment 9: Windows Exploitation using Metasploit (TryHackMe)

Aim

To exploit Windows OS vulnerabilities using Metasploit Framework in TryHackMe.

Procedure

1. Start Windows exploitation room.
2. Launch AttackBox.
3. Open Metasploit.
4. Configure exploit.
5. Execute exploit.
6. Gain Meterpreter session.

Commands Used

```
msfconsole  
search windows  
use exploit/windows/...  
set RHOST <IP>  
set LHOST <IP>  
exploit
```

Output

Meterpreter session established.

Result

Thus, Windows exploits were demonstrated using Metasploit in TryHackMe.

Experiment 10: Wireless Audit and WPA Cracking using Aircrack-ng (TryHackMe)

Aim

To perform wireless auditing and WPA key cracking using Aircrack-ng in TryHackMe.

Procedure

1. Start wireless security room.
2. Launch AttackBox.
3. Enable monitor mode.
4. Capture WPA handshake.
5. Perform deauthentication.
6. Crack WPA key.

Commands Used

```
airmon-ng start wlan0  
airodump-ng wlan0mon  
aireplay-ng --deauth 5 -a <BSSID> wlan0mon  
aircrack-ng capture.cap -w wordlist.txt
```

Output

WPA key decrypted successfully.

Result

Thus, wireless audit and WPA cracking were demonstrated in TryHackMe.